Retrieval-Augmented Generation for AI-Powered Multimodal Knowledge Intelligence Systems

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Abstract

Large language models (LLMs) are prone to hallucination and rely on static, time-bound training data, which limits their reliability for knowledge-intensive tasks. Retrieval-Augmented Generation (RAG) addresses this by grounding a model's output in evidence retrieved from an external corpus at inference time. This paper reviews the foundations of RAG and its extension to the multimodal setting, in which the retrieval corpus and the query itself may span text, images, audio, and video. We present a proposed methodology for our project, the AI-Powered Multimodal Retrieval and Knowledge Intelligence System, covering multimodal data ingestion, embedding and indexing, hybrid retrieval, cross-modal fusion, and grounded response generation with source attribution. We also outline an evaluation plan based on retrieval quality metrics and generation quality metrics, and discuss the practical and research challenges of building a reliable multimodal RAG pipeline, including cross-modal alignment, latency, and evaluation.

Keywords: RETRIEVAL-AUGMENTED GENERATION, MULTIMODAL RAG, VECTOR DATABASE, LARGE LANGUAGE MODELS, KNOWLEDGE INTELLIGENCE, SEMANTIC SEARCH, EMBEDDINGS, CROSS-MODAL RETRIEVAL.

1 Introduction

Large language models encode a broad amount of world knowledge in their parameters, but this knowledge is frozen at the point of training, cannot be easily updated, and is not traceable back to a verifiable source. When asked about recent events, private organisational documents, or long-tail facts, an LLM will often produce a fluent but incorrect answer — a failure mode commonly referred to as hallucination. Retrieval-Augmented Generation (RAG), introduced by Lewis et al. [1], addresses this problem by coupling a parametric language model with a non-parametric retriever: at inference time, the system first retrieves the passages most relevant to the query from an external corpus and then conditions the language model's generation on that retrieved evidence, rather than relying solely on memorised parameters.

Classical RAG operates over text alone. However, real-world knowledge bases — technical manuals, lecture recordings, product catalogues, scanned reports — are inherently multimodal. Multimodal Retrieval-Augmented Generation (MRAG) extends the RAG paradigm so that both the corpus and the query can include text, images, audio, and video, and so that retrieval and generation can reason jointly across these modalities [2]-[4]. This paper reviews the state of MRAG and describes how it is applied in our project, an AI-Powered Multimodal Retrieval and Knowledge Intelligence System that is intended to answer questions grounded in mixed-modality organisational knowledge.

2 Related Works

Lewis et al. [1] proposed the original RAG framework, combining a dense passage retriever with a sequence-to-sequence generator and showing that grounding generation in retrieved passages improves factual accuracy on knowledge-intensive NLP tasks compared to purely parametric models. Follow-up work explored retrieval-augmented pretraining and in-context retrieval, establishing RAG as a standard technique for reducing hallucination in production LLM systems.

More recently, several surveys have mapped the extension of RAG to multiple modalities. Zhao et al. [2] survey methods for retrieving multimodal information to augment generation, covering applications in factuality, reasoning, interpretability, and robustness. Mei et al. [3] present a survey of multimodal RAG that reviews essential components, datasets, and evaluation methods, and identify open challenges and future directions for the field. Abootorabi et al. [4] provide a comprehensive survey of multimodal RAG systems, covering datasets, benchmarks, evaluation metrics, and innovations across retrieval, fusion, augmentation, and generation, and further discuss training strategies, robustness enhancements, and agent-based multimodal RAG approaches. A common theme across this literature is that cross-modal alignment — ensuring that a text query can

correctly retrieve relevant images, audio, or video, and vice versa — introduces challenges beyond those found in unimodal, text-only RAG [2]-[4].

3 Proposed Methodology

We propose a multimodal RAG pipeline for the AI-Powered Multimodal Retrieval and Knowledge Intelligence System, structured as six stages: data collection and pre-processing, multimodal embedding and indexing, query encoding, hybrid retrieval, cross-modal fusion, and grounded generation. Figure 1 depicts the overall system architecture.

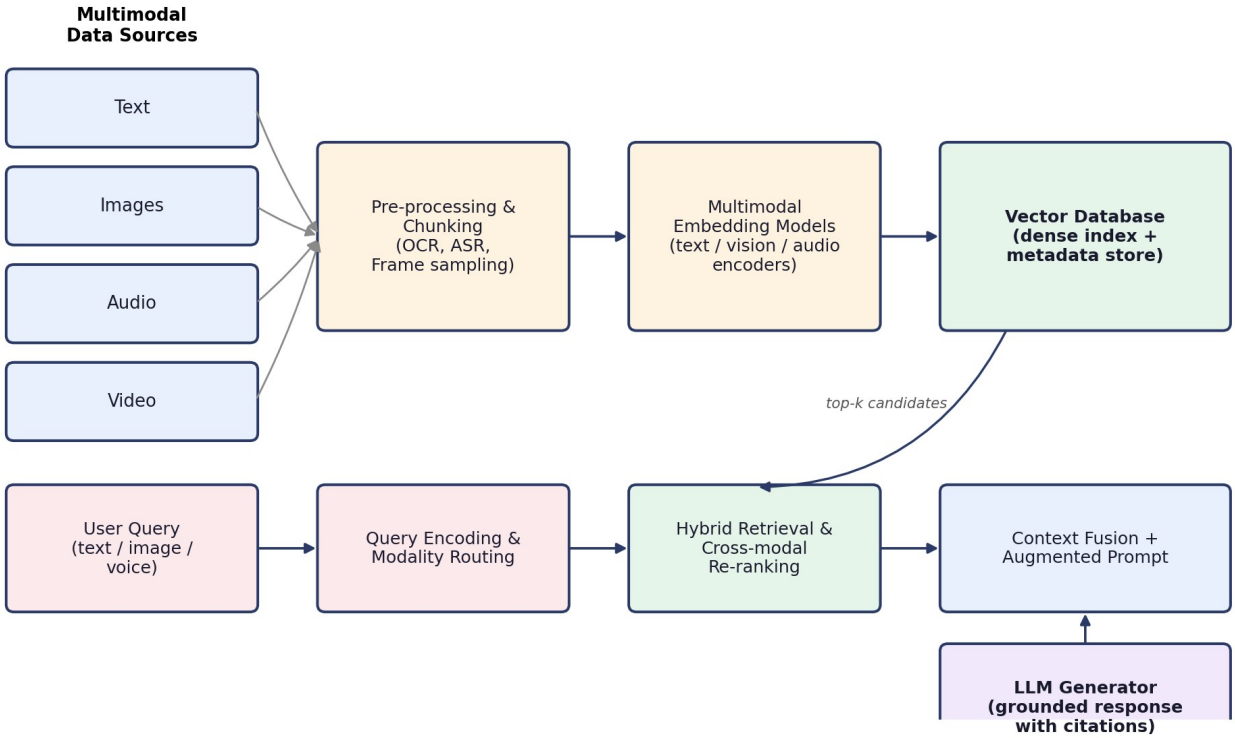


Fig. 1: Proposed multimodal retrieval-augmented generation architecture

3.1 Data Collection and Pre-processing

Source documents are collected from heterogeneous formats — PDFs, slide decks, scanned images, audio recordings, and video lectures. Text is extracted directly where available and through optical character recognition (OCR) otherwise; audio and the audio track of video are transcribed using an automatic speech recognition (ASR) model; video is additionally sampled into representative frames. Each modality is then segmented into retrieval-sized units: text is chunked by paragraph or sliding window, images are treated as individual units, and audio/video are chunked by time interval, with metadata (source, page/timestamp, modality) preserved for later citation.

3.2 Multimodal Embedding and Indexing

Each chunk is projected into a shared embedding space using modality-appropriate encoders: a text embedding model for text chunks, a vision-language encoder for images and video frames, and a text embedding model applied to ASR transcripts for audio. Embeddings, together with their metadata, are stored in a vector database that supports approximate nearest-neighbour search alongside metadata filtering, allowing the retriever to narrow results by source type, date, or document.

3.3 Query Encoding and Retrieval

An incoming query — which may itself be text, an image, or a spoken question — is encoded with the same family of encoders used for indexing, and a hybrid retrieval strategy combines dense vector similarity with sparse keyword matching to improve recall on exact terms such as product codes or names that dense embeddings can under-represent. The top-k candidates across all relevant modalities are retrieved.

3.4 Cross-Modal Fusion and Re-ranking
Retrieved candidates from different modalities are re-ranked jointly using a cross-encoder that scores the relevance of each candidate to the query, and redundant or near-duplicate evidence is filtered out. The surviving evidence set is then formatted into a structured, source-attributed context block that will be passed to the generator.

3.5 Grounded Generation
The language model receives the user query together with the fused evidence context and is prompted to produce an answer that is explicitly grounded in, and cites, the retrieved sources. Where the retrieved evidence is insufficient or contradictory, the system is designed to state this explicitly rather than fabricate an answer, following the general RAG principle that generation should be conditioned on, and constrained by, retrieved evidence [1].

3.6 Proposed Algorithm
Input: Multimodal knowledge corpus (text, images, audio, video) and a user query.
Output: A grounded, source-attributed answer to the query.
- 1. Ingest and pre-process the corpus; extract text, transcribe audio/video, and sample video frames.
- 2. Chunk each modality and compute embeddings; store embeddings and metadata in the vector index.
- 3. Encode the incoming query using the matching modality encoder(s).
- 4. Retrieve top-k candidates per modality using hybrid dense + sparse search.
- 5. Re-rank and fuse the candidates across modalities; remove redundant evidence.
- 6. Construct an augmented prompt containing the query and the fused, cited evidence.
- 7. Generate a grounded answer with the LLM, including citations to source chunks.

8. Return the answer to the user; log the query and evidence trail for evaluation.

4 Evaluation Plan and Discussion
Because this paper presents the proposed design of the retrieval-augmented pipeline rather than results from a completed deployment, we outline the evaluation methodology that will be used to validate the system rather than reporting specific performance figures. Retrieval quality will be measured using standard information-retrieval metrics such as Precision@k, Recall@k, and normalised discounted cumulative gain (nDCG), computed separately for each modality and for cross-modal queries. Generation quality will be assessed with a combination of automatic metrics — such as ROUGE and BERTScore against reference answers — and human evaluation of factual groundedness, i.e., whether every claim in the generated answer is supported by a cited source. System-level metrics such as end-to-end latency and retrieval-index build time will also be tracked, since multimodal encoders and cross-encoder re-ranking are computationally heavier than text-only retrieval.
We expect the main practical challenges to mirror those reported in the multimodal RAG literature [2]-[4]: aligning embeddings across modalities so that a text query reliably surfaces relevant images or audio segments; balancing retrieval recall against latency as the corpus grows; and preventing the generator from over-relying on a single dominant modality when evidence conflicts across sources. Addressing these challenges will be the focus of the implementation and testing phase of the project.

5 Conclusion and Future Work
Retrieval-Augmented Generation grounds language model output in external, verifiable evidence and is an effective remedy for hallucination and stale knowledge in knowledge-intensive applications. Extending RAG to the multimodal setting, as required by our AI-Powered Multimodal Retrieval and Knowledge Intelligence System, introduces the additional challenge of aligning and jointly ranking evidence drawn from text, images, audio, and video. We proposed a six-stage pipeline covering ingestion, multimodal embedding and indexing, hybrid retrieval, cross-modal fusion, and grounded generation, and outlined an evaluation plan combining retrieval metrics, generation-quality metrics, and system-level latency measurements. Future work will implement and benchmark this pipeline against a text-only RAG baseline, and, building on the agentic architecture discussed in our companion paper on Agentic AI, explore an agent-driven controller that plans multi-hop retrieval across modalities rather than issuing a single retrieval call per query.

6 References

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